

Build Approach and Considerations

How the prototype was approached, what was prioritized, and what tradeoffs were made under the time constraint.

Approach

- I prioritized the executive demo story first: a marketing manager should move from connected retail data to an exportable campaign package in one session.
- The first screen is the working product rather than a landing page. The prototype needed to prove the workflow, not just describe it.
- The main build priority was an honest end-to-end loop: real frontend, real backend, structured data, generated assets, explainable scoring, and practical export behavior.

Workflow Priorities

- Load campaign history, product catalog data, and customer segment profiles.
- Surface a current recommendation and let the marketer apply it or adjust the brief.
- Use LangGraph to analyze performance patterns, generate channel variants, and rank them by projected CTR, ROAS, lift, confidence, and audience fit.
- Keep live AI actions optional and scoped to recommendation refresh, brief refinement, copy polish, selected images, and one selected TikTok video.
- Export the selected campaign state in the same shape that could later be stored in S3.

Technology Choices

Technology	Reasoning
FastAPI	Small, typed Python API for demo endpoints, static files, health checks, media adapters, and exports.
LangGraph	Natural fit for the load, analyze, recommend, generate, score, and refine workflow.
React + Vite + TypeScript	Fast iteration and type-safe UI state for a polished single-page campaign console.
TanStack Query	Clean handling for user-triggered generation, media, recommendation, brief, and export mutations.
Pydantic settings	Centralized provider, model, CORS, and cost-control configuration.
Provider adapters	OpenAI, GPT Image 2, Seedance, Bedrock, and Nova Reel paths stay swappable behind backend interfaces.
Local ZIP export	Immediate campaign handoff today, with a direct path to S3, metadata storage, and presigned URLs later.

Tradeoffs

- The performance model is heuristic rather than trained ML; it computes real metrics from synthetic data and applies rules for forecasted lift and fit.
- Product and ad assets are generated demo references instead of real retailer photography, which keeps the prototype self-contained and privacy-safe.
- Image generation is limited to selected top previews, and live video targets one TikTok creative, to control cost and latency.
- The prototype creates a campaign handoff package but does not publish directly into paid-media or email platforms.
- The CloudFormation file is a deployment skeleton rather than a fully hardened production stack.

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Continuation: implementation challenges, future work, and delivery notes.

Interesting Challenges

- Keeping the prototype honest: the UI calls a backend endpoint, the backend runs a LangGraph workflow, and outputs change based on products, segments, channels, objective, and strategy.
- Managing cost: the default path is deterministic and local, while live AI is opt-in and limited to specific actions.
- Making agent behavior understandable: the UI uses marketer-facing language such as recommendations, brief, ranked previews, and rationale, while technical details stay in docs and code.

What I Would Do With More Time

- Connect real S3 product assets, campaign exports, and analytics data.
- Train or calibrate a creative-response model with historical performance data.
- Add brand safety, legal review, approval states, audit logs, and creative memory.
- Extend image/video generation to controlled batch generation with human review.
- Add direct publishing integrations for paid social, display, search, and email platforms.
- Harden the AWS stack with CI/CD, observability, private networking, least-privilege IAM, and environment-specific configuration.

Delivery Notes

- README includes setup, run, verification, environment, export, and cost-control instructions.
- Architecture and build approach PDFs are generated into docs/ for direct submission.
- The repo includes a one-command dev runner plus a PDF generation script for reproducible documentation.